

The BIND Lightcone Suite III: The Two-Dimensional Latent Space of Baryonic Feedback, and What Lensing $\times$ SZ Statistics Can Constrain

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Abstract

Marginalizing baryonic feedback in Stage-IV weak-lensing (WL) analyses requires knowing the *effective dimensionality* of the feedback response, not merely its amplitude. Using BIND-painted lightcones over a fixed-cosmology, 30-parameter (SB35) Sobol sweep of the IllustrisTNG feedback prior, we show that the WL convergence power-spectrum suppression $S(\ell)$ collapses onto a two-dimensional latent space: linear PCA on a 216-run subset of the sweep explains $0.884 + 0.103 = 98.7\%$ of the suppression variance in two components, and an independent β -TC-VAE decomposition agrees. A repeat of this measurement on the final 253-run sweep gives a consistent PC1 $\approx 89\%$, PC2 $\approx 11\%$ (bar heights read independently off the figure and individually rounded, so they do not sum exactly to the cumulative below), cumulative 99.2%. We flag an important, unresolved tension with this headline: a related field-level construction that additionally folds in the $\kappa \times y$ cross-spectrum finds a lower effective dimensionality ($\lambda_1 \approx 0.95$ for a single leading mode); see §3.1 for the caveat and what is and is not yet reconciled between the two measurements. The two axes have a clean physical reading: one tracks a black-hole (quasar/radio) feedback direction, the other a supernova/wind direction. This independently reproduces the two-dimensional feedback latent recently reported for the hydrodynamic-to-DMO power-spectrum ratio across CAMELS suites (Lin et al., 2026). Canonical-correlation analysis shows the leading WL latent axis is nearly identical to the leading kinetic-SZ (kSZ) gas latent axis (canonical correlation 0.98, principal angle 12°), while the second axis is substantially rotated (0.79, 38°): WL lensing responds to a blend of the inner and outer gas zones that kSZ separates cleanly ($r = 0.59/0.71$ for WL versus $r \approx 0.95$ for kSZ). A model-free (emulator-independent) Spearman response map over the final $n = 253$ suite ranks the IMF slope, the wind-velocity factor, the black-hole radiative efficiency, and the wind energy as the top drivers of $S(\ell)$ ($|r| = 0.49, 0.48, 0.43, 0.34$). Simulation-based inference (SBI) on the WL auto-spectrum alone leaves the 30-parameter posterior broad (only ~ 5 of 30 parameters shrink appreciably below the prior), yet the same posterior projected onto the 2-D suppression latent is tight, spanning only ~ 0.17 – 0.18 of the prior-predictive node cloud, and passes simulation-based calibration (mean 68% coverage = 0.70). A cautionary methods result: naively stacking emulated higher-order statistics and the $\kappa \times y$ cross-spectrum onto the auto- C_ℓ likelihood *degrades* rather than improves the fit to the measured fiducial (mean posterior shrink $0.97 \rightarrow 1.01 \rightarrow 1.03$), because the emulator’s fidelity for these statistics is lower than for $S(\ell)$ itself, not because the extra statistics are uninformative. Consistent with this, a field-level, emulator-free analysis of the same suite finds that the $\kappa \times y$ cross-spectrum and tSZ auto-spectrum, not the WL auto-spectrum, carry most of the field-level feedback information, pinning two directions to variance 0.06/0.11 at fixed cosmology. Using the actual lightcone halo population (not a single low-redshift snapshot), the per-halo Born-approximation lensing perturbation correlates with halo gas fraction at $r = +0.62$,

peaking near $z \approx 0.33$, even though the low-redshift shell used in earlier work carries only $\sim 3\%$ of the source-plane lensing-kernel weight. Finally, we present the first direct confrontation of BIND shear $\times y$ against real DES Y3 $\times$ ACT data: BIND, painted at the fiducial TNG point, over-predicts the correlation (WORKLOG’s rounded characterization is $\sim 1.5\text{--}2\times$; the precise robust-aperture readout is $\approx 2.3\text{--}2.5\times$, §3.7) and is too steep in angle, indicating IllustrisTNG feedback is too gas-bound relative to the observed Universe at this level of analysis. **[TODO: A population-level SBI posterior on the ~ 8.6 million per-halo $(\theta, \log M_{200c}, a, \log Y_{200})$ tuples was designed and partially built (amortized NPE + IID-product likelihood + leave-one-run-out temperature calibration) but has not been run to a completed GPU result; we report the method here and defer the calibrated population posterior to future work.]**

1 Introduction

Weak gravitational lensing (WL) by large-scale structure is one of the primary cosmological probes of the next decade, but its statistical power is fundamentally limited by our ignorance of how baryonic feedback redistributes matter relative to a dark-matter-only (DMO) universe. Feedback from active galactic nuclei (AGN) and supernova-driven winds evacuates gas from halos and suppresses small-scale power in a way that, if unmodeled, biases cosmological parameter inference from cosmic shear at the several- σ level for Stage-IV surveys such as LSST and Euclid (van Daalen et al., 2020; Schneider et al., 2022; Amon et al., 2022). The standard mitigation strategies are either to marginalize over phenomenological baryon models (baryonification / BCM-style parameters; Aricò and Angulo, 2024; Schneider et al., 2022) or to build large hydrodynamic simulation suites and directly emulate or marginalize the response (Villaescusa-Navarro et al., 2021; Schaye et al., 2023). Both strategies implicitly assume the response has a large number of independent nuisance directions, but the actual dimensionality of the astrophysical feedback response has rarely been measured directly. A companion paper in this suite finds that exactly two data-driven baryon templates are necessary and sufficient to remove the IllustrisTNG WL baryonic bias from a mock LSST-Y10 cosmic-shear analysis (Lee, 2026b, Paper II); this paper asks the more basic question the capstone result presupposes — is the feedback response to a high-dimensional astrophysical prior itself low-dimensional, and if so, what physical directions does it correspond to, and what field-level statistics carry that information?

A parallel finding has emerged in the CAMELS literature: the ratio of the hydrodynamic to DMO matter power spectrum, $T^2(k, a) = P_{\text{hydro}}/P_{\text{DMO}}$, computed across many CAMELS simulation suites and astrophysical parameter combinations, is well described by a two-dimensional latent space (Lin et al., 2026). This paper performs the direct WL analogue of that measurement — using the field-level BIND emulator painted onto IllustrisTNG dark-matter-only lightcones rather than the 3-D matter power spectrum — and extends it in three directions relevant to observational cosmology: (i) we ask whether the same 2-D latent space is shared with an independent gas-thermodynamic probe, the kinetic Sunyaev–Zel’dovich (kSZ) effect, using canonical-correlation analysis; (ii) we quantify, with simulation-based inference (SBI), how tightly realistic field-level statistics actually pin that latent space, as distinct from the full 30-parameter astrophysical prior; and (iii) we confront the resulting model directly with real cross-correlation data.

Recovering feedback parameters (or a low-dimensional feedback summary) from field-level statistics is naturally framed as a simulation-based (or likelihood-free) inference problem, since the forward map from astrophysical parameters to lensing and SZ observables has no tractable likelihood (Cranmer et al., 2020). We use neural posterior estimation (NPE), specifically sequential/amortized normalizing-flow posterior estimators in the spirit of Papamakarios and Murray (2016) and the automatic posterior transformation of Greenberg et al. (2019), as implemented in the `sbi` package

(Tejero-Cantero et al., 2020), and we validate calibration with simulation-based calibration (SBC) rank statistics (Talts et al., 2018). Because a 30-parameter Sobol design of a few hundred simulations is itself data-starved for flow-based NPE, we additionally use the BIND field-statistics emulator (Lee, 2026a, Paper I, in prep) as an unlimited stochastic simulator with calibrated noise, which lets us probe the SBI question at a level of precision the raw simulation count alone would not support — with an important caveat about emulator fidelity that we return to in Section 4.

On the observational side, cross-correlations of galaxy shear with the thermal SZ (tSZ) effect and with the kSZ effect are increasingly used to break the lensing-amplitude/feedback degeneracy that limits WL auto-power alone (Hill and Spergel, 2014; Battaglia et al., 2012; Amodeo et al., 2021; Tröster et al., 2022; Gatti et al., 2022b; Pandey et al., 2022, 2025; Bigwood et al., 2024), and higher-order WL statistics (peaks, Minkowski functionals, scattering transforms) carry feedback information beyond the two-point function (Martinet et al., 2021; Zürcher et al., 2021; Cheng et al., 2020; Gatti et al., 2022a, 2024). This paper asks which of these statistics actually carry the feedback information in a field-level, simulation-based sense, using the same BIND lightcone suite throughout so that every probe is evaluated on identical sky realizations and the same underlying halo population.

The remainder of the paper is organized as follows. Section 2 describes the Sobol simulation suite, the statistics vector, the two independent latent-extraction methods, the canonical-correlation and SBI machinery, and the model-free global-response diagnostic. Section 3 presents seven results: the 2-D latent space and its physical axes (§3.1); the WL↔kSZ latent geometry (§3.2); a model-free global-response map (§3.3); SBI on the WL auto-spectrum, including a cautionary result about combining statistics of unequal emulator fidelity (§3.4); the dominance of $\kappa \times y$ and tSZ in field-level feedback information (§3.5); a halo-level lensing/gas connection validated on the actual lightcone halo population (§3.6); and a first direct confrontation with real DES Y3 $\times$ ACT shear $\times y$ data (§3.7). Section 4 discusses what the 2-D result implies for feedback marginalization, the emulator-fidelity bottleneck exposed by the SBI cautionary result, and the TNG-only/fixed-cosmology limitations of the whole analysis. An optional appendix (§A) reports a related finding on the stellar-mass–halo-mass relation (SHMR) scatter budget from the same Sobol suite.

2 Methods

2.1 Suite recap and design-size bookkeeping

The underlying suite is described in full in Papers I and II (in prep); we recap the elements relevant here. BIND paints hydrodynamic fields onto IllustrisTNG dark-matter-only lightcones conditioned on a 30-dimensional astrophysical parameter vector drawn from the CAMELS SB35 prior, at fixed (TNG) cosmology. The analysis in this paper uses a single 256-node Sobol design over that 30-parameter prior, with every design point painting the *same* shared, fiducial dark-matter halo population, so that run-to-run differences in any downstream statistic are attributable to the astrophysical parameters alone rather than to sample variance in the underlying structure.

Because this suite was analyzed as it grew, different results below use different completed-run counts N ; we state N explicitly with every number and summarize the bookkeeping in Table 1.

2.2 Statistics vector

For every completed Sobol run we compute, at up to five tomographic source planes, the WL convergence auto-spectrum $C_\ell^{\kappa\kappa}$, the $\kappa \times y$ cross-spectrum, the tSZ auto-spectrum C_ℓ^{yy} , and the kSZ optical-depth auto-spectrum $C_\ell^{\tau\tau}$; where a DMO-normalized ratio is meaningful we report the

Table 1: Growth of the analyzed Sobol sample size over the course of the project (all draws from the same 256-node design). Every quantitative result in §3 is labeled with the N used. [†]The §3.5 engines hardcode $N = 123$ in their own docstrings; we were not able to confirm from the mined sources whether this was re-run at the final $N = 253$, so we report the number as documented and flag it ([TODO: confirm or re-run the $\kappa \times y$ /tSZ multiprobe analysis at $N = 253$]).

N	Date	Analysis
123	2026-06-23	first NPE pass on the raw C_ℓ (<code>lightcone_cl_sbi.py</code>)
123 [†]	2026-06-24	$\kappa \times y$ /tSZ field multiprobe (<code>wl_sz_multiprobe.py</code> , <code>wl_sz_cosmo_anchor.py</code> ; §3.5)
216	2026-06-24	auto- C_ℓ NPE refresh; WL feedback-latent β -TC-VAE
237	2026-06-24	<code>bind.emulator</code> full-set rebuild
253	2026-06-29	final state; <code>wl_latent_sbi.ipynb</code> / <code>paper_lightcone_figs2.ipynb</code>

suppression $S(\ell) = C_\ell^{\kappa\kappa}/C_{\ell,\text{DMO}}^{\kappa\kappa}$, which cancels cosmic variance because BIND and its paired DMO trace share the same underlying structure. We additionally compute peak and minima counts and the one-point PDF of the convergence field, the three Minkowski functionals (V_0, V_1, V_2), a wavelet scattering transform (WST; Cheng et al., 2020, via `kymatio`), and binned scaling relations ($Y-M$, $f_{\text{gas}}-M$, $T-M$). The WST requires a compatibility shim for `scipy` ≥ 1.13 and had only 40 valid runs computed at the time of the cross-statistic latent comparison in §3.2, so it is excluded from that comparison; we flag this explicitly as a caveat rather than silently omitting it.

2.3 Latent extraction

We extract a low-dimensional feedback latent space from $\log_{10} S(\ell, z_s)$ (or another binned statistic) using two independent methods that we find agree (§3.1). The first is linear PCA truncated at two components. The second is a conditional β -total-correlation VAE (β -TC-VAE; the total-correlation decomposition of Chen et al., 2018), trained full-batch with KL annealing, which achieves 1.3% RMSE on the reconstructed suppression field with a 2-dimensional latent bottleneck. We further cross-check the effective dimensionality against a nonlinear information-ordered bottleneck (an MLP trained to maximize an information-curve $R^2(k)$ as a function of latent rank k) and an active-subspace decomposition (the eigendecomposition of $\mathbb{E}[J^\top J]$ for the Jacobian of a $\theta \rightarrow x$ surrogate); the nonlinear information curve plateaus by the second latent rank, consistent with the linear result.

2.4 Canonical correlation between WL and kSZ/gas latents

A note on terminology. Throughout this paper “kSZ” denotes BIND’s velocity-free proxy for the kinetic Sunyaev–Zel’dovich effect — the electron-column optical depth τ and thermal-pressure y radial profiles — not a genuine velocity-weighted kSZ temperature-decrement signal: BIND has no gas-velocity field, so a true ΔT_{kSZ} map is not yet constructed. Concretely, the “kSZ latent” used below (`ksz_latent` in `wl_latent_sbi.py`) is built purely from stacked $\log_{10} \tau$ and $\log_{10} y$ radial profiles. To test whether the WL latent space is the same latent space accessible to gas-thermodynamic probes, we compute the standard canonical-correlation analysis (CCA; Hotelling, 1936) between the top-2 PCA latent of each field statistic and the top-2 PCA latent of $C_\ell^{\kappa\kappa}$, together with the principal angle between the corresponding 2-D subspaces (`align_planes` in `wl_latent_sbi.py`, using `sklearn.cross_decomposition.CCA`). This reuses the kSZ τ/y latent kernel from the sibling kSZ analysis (Lee, 2026c, Paper IV, in prep).

2.5 Simulation-based inference

We use two generations of NPE. The first runs directly on the measured Sobol simulations ($N = 123 \rightarrow 216$; `lightcone_cl_sbi.py`), which is data-starved at these sample sizes: we work in a bounded $[0, 1]^{30}$ prior transformed to logit space (naive rejection sampling in the bounded box has acceptance $\sim 0.8^{30} \approx 10^{-3}$ and stalls), and pool a 5-flow normalizing-flow ensemble for stability, since a single flow’s posterior shrink for the wind-velocity parameter varies between 0.97 and 1.12 across random-seed draws. The second generation (`wl_latent_sbi.py` §3) uses the `bind.emulator` Gaussian-process surrogate as an `EmulatorSimulator` — an effectively unlimited ($\theta \rightarrow x$) generator with noise calibrated to the emulator’s own residuals — which escapes the $N = 253$ data-starvation regime that limits the first generation. We run NPE against this simulator, cross-check the resulting posterior against an explicit-likelihood `emcee` fit, and validate calibration with SBC rank histograms (Talts et al., 2018). In all cases the inference target is the *measured* BIND fiducial realization (known ground-truth θ).

2.6 Per-halo population SBI

In addition to the field-level analysis above, we designed an amortized, per-halo population-inference pipeline: a neural posterior estimator $q(\theta \mid x_{\text{halo}})$ trained on ~ 8.6 million per-halo $(\theta, \log M_{200c}, a, \log Y_{200})$ tuples drawn from the same painted DMO halos across the 256 design points and 20 lightcone redshift shells. The population posterior is formed from the IID product $\log p(\theta \mid \{x_i\}) = \sum_i \log q(\theta \mid x_i) + \text{const}$ under a flat prior on the unit box, sampled in 30 dimensions with `emcee`; honesty is checked with leave-one-run-out coverage, with a likelihood “temperature” T (an effective $1/N_{\text{eff}}$) calibrated so that the leave-out 68% credible band actually covers 68% of held-out truths. [TODO: This pipeline was designed and its engines exist (`sobol_perhalo_sbi.py`, `train_perhalo_sbi.py`), but it was not run to completion on GPU: the executed notebook explicitly reports no cached GPU results and falls back to a small inline model labeled “illustration only, not the science result,” and no `population_posterior.npz` exists on disk. We therefore do not report a population-level posterior in §3; the method is documented here for completeness and as a specific, well-defined item of future work.]

2.7 Model-free global-response maps

Because the emulator used for SBI (§2.5) is validated to be accurate for the suppression *amplitude* but is not trusted to rank fine sensitivities, we additionally compute a deliberately emulator-free diagnostic: the per-bin Spearman rank correlation between each measured field statistic and each of the 30 astrophysical parameters, evaluated directly on the $N = 253$ measured Sobol runs (`bin_param_correlation` in `wl_latent_sbi.py` §1). An earlier, emulator-pushed Sobol variance-decomposition (`global_sensitivity/sobol_indices`) exists in the same codebase but is explicitly deprecated by its own docstring in favor of the Spearman map; we discuss the disagreement between the two in §3.3.

3 Results

3.1 The 2-D latent space of the WL suppression

Linear PCA on $\log_{10} S(\ell, z_s)$ across the $N = 216$ Sobol subsample explains 0.884 and 0.103 of the variance in its first two components, a cumulative 98.7%; the β -TC-VAE recovers a consistent two-dimensional latent (Figure 1a). An independent repeat of the same PCA measurement on

the final $N = 253$ sweep, embedded directly in the paper figure notebook, gives $\text{PC1} \approx 89\%$, $\text{PC2} \approx 11\%$, cumulative 99.2% (Figure 3) — consistent with, though not numerically identical to, the 0.884/0.103 figure above, reflecting a different exact ℓ -cut and binning between the two measurements. We adopt the 0.884/0.103 (98.7%) figure as the headline number and treat the 99.2% measurement as an independent consistency check rather than a correction.

An unresolved tension with a companion field-level measurement. A sibling notebook analyzing the same $N = 253$ suite, `paper_ksz_field.ipynb` (WORKLOG 2026-06-26; also the source of Figures 11 and 13 below, §3.5 and §3.7), performs a closely related PCA: it stacks $\log_{10} S(\ell, z_s)$ over the same five tomographic source planes *together with* the $\kappa \times y$ cross-spectrum (mean-subtracted, linear, not log-transformed), band-limited to $\ell \in (300, 8000)$, versus our headline construction (`wl_suppression_stack` in `wl_latent_sbi.py`), which stacks $\log_{10} S(\ell, z_s)$ *alone* over $\ell \in [100, 2 \times 10^4]$. That notebook reports the field response is “*essentially 1-dimensional*”: one mode carries $\lambda_1 \approx 0.95$ of the variance ($\lambda_2 \approx 0.04$, cumulative-at-2 = 0.99), and states this is “robust to the probe combination” including a $\kappa \times y$ -excluded, WL-suppression-only ablation, which it reports also gives $\lambda_1 \gtrsim 0.93$ — noticeably higher than our own $\text{PC1} = 88.4\text{--}89\%$ on what is, at least nominally, a similar tomographic $S(\ell, z_s)$ stack. We have identified real construction differences between the two measurements (the ℓ -band, the treatment of $\kappa \times y$, and linear versus log normalization of the cross-spectrum blocks) but have not been able to reproduce the companion notebook’s ablation ourselves — its “WL-suppression-alone” code path is not present in the mined sources, only asserted in prose — so we cannot yet say whether these differences fully account for the gap between 88–89% and $\gtrsim 93\text{--}95\%$ for the leading mode, or whether the two notebooks’ pipelines have a deeper, unreconciled inconsistency. [TODO: Run both PCA constructions on an identical, explicitly shared input matrix (same ℓ -band, same run list, same $\kappa \times y$ treatment) to determine whether the 2-D/ $\sim$ 1-D difference is a construction artifact or a genuine result.] We note that our own 2-D claim does not rest on this single PCA: it is corroborated independently by the β -TC-VAE reconstruction below, the $N = 253$ repeat measurement above, the nonlinear information-bottleneck plateau at rank 2 (§2.3), and the two visibly distinct 1P AGN/SN spoke directions in Figure 3; conversely, we cannot rule out that the companion notebook’s inclusion of $\kappa \times y$ — which §3.5 independently shows to be strongly amplitude-dominated — is pulling its leading SVD mode toward a single amplitude direction in a way that is physically real for *that* statistic combination without contradicting the $S(\ell, z_s)$ -only result reported here. We leave a first-principles reconciliation to future work and flag this explicitly rather than letting the two internally-cited results stand in silent tension.

Figure 2 shows this physical-correlate readout in full: eleven panels, each the same (latent-1, latent-2) plane colored by a different physical quantity, confirming that latent 2 in particular tracks the gas-thermodynamic quantities directly and not merely as an artifact of the WL suppression construction. The same $N = 253$ suite, viewed directly on the (latent-1, latent-2) plane rather than colored by physical quantities, shows the Sobol nodes fan out around the fiducial point along the two directions identified above, with the astrophysical one-parameter-at-a-time (1P) AGN and SN/wind spokes aligned with the two axes as expected (Figure 3).

3.2 WL $\leftrightarrow$ kSZ latent geometry

The CCA between the WL suppression latent and the kSZ τ/y latent (both from the same $N = 253$ suite) gives canonical correlations of [0.98, 0.79] and principal angles of [12°, 38°] between the two 2-D planes (Figure 4). The dominant axis is essentially shared between the two probes, while the second axis is rotated by nearly 40°: WL convergence, being a line-of-sight-projected total-mass perturbation, responds to a *blend* of the inner and outer gas zones that kSZ’s τ/y decomposition

separates cleanly, tracking the inner and outer f_{gas} zones at only $r = 0.59$ and 0.71 respectively, versus $r \approx 0.95$ for the corresponding kSZ latent.

To test whether this rotation is peculiar to the auto- C_ℓ or is a generic property of WL summary statistics, we align every WL summary’s own top-2 PCA latent to the κ - C_ℓ 2-D latent. The first latent axis is nearly universal across statistics (alignment ≥ 0.93 for all eight tested: peak counts 0.99, minima counts 0.99, PDF 1.00, Minkowski V_0 0.98, V_1 1.00, V_2 1.00, $C_\ell^{\kappa y}$ 0.97, C_ℓ^{yy} 0.93), but the second axis alignment is strongly statistic-dependent: it is high for peaks/PDF/minima (0.69–0.76), low for the Minkowski functionals (0.20–0.43), and *highest* of all for the tSZ cross- and auto-spectra (0.90, 0.89; Figure 5). This is consistent with tSZ probes seeing the same rotated second direction that the kSZ latent sees, and with the topology and higher-order Minkowski statistics seeing a different (more mixed) combination.

Orthogonal tSZ information does not expand the number of astrophysical parameters recoverable from the field (cross-validated ridge $R^2 > 0.1$ stays at $\sim 5/30$ for κ two-point statistics alone, all-WL, or WL+tSZ combined), but it does sharpen and rotate which of those five parameters are accessible: IMF-slope recoverability rises from 0.33 (κ alone) to 0.56 (all-WL) to 0.64 (WL+tSZ), and tSZ specifically unlocks the thermal-energy/wind sector — wind-energy recoverability rises from 0.13 to 0.33, and radio-feedback reorienting becomes recoverable at 0.16 only once tSZ is included.

3.3 Global response: which parameters actually drive $S(\ell)$

The model-free (emulator-independent) Spearman response map over the $N = 253$ measured suite (Figure 6) ranks the IMF slope ($|r| = 0.49$), the variable wind-velocity factor (0.48), the black-hole radiative efficiency (0.43), and the wind energy (0.34) as the top drivers of $S(\ell, z_s)$ at $z_s = 1$; peak-statistic sensitivity is concentrated in the high- ν tail of the peak function.

We caution that a second, emulator-pushed global-sensitivity metric exists in the same codebase (a Sobol variance decomposition, `global_sensitivity/sobol_indices`), and it *disagrees* with the Spearman ranking above: it ranks the wind-velocity factor as the dominant driver ($S_i \approx 0.33$) and the black-hole radiative efficiency second ($S_i \approx 0.23$), and does not surface the IMF slope as a top driver at all. The docstring of the code that produces this second metric explicitly states it is “kept for reference but no longer drives” the headline response analysis, and we follow that judgment: we report the Spearman map as the primary result and flag the disagreement rather than silently choosing one ranking. [TODO: Reconcile or explain the Sobol-index vs. Spearman disagreement quantitatively in a future revision; at present we can only report that they disagree on the top driver.]

3.4 Simulation-based inference on the WL auto-spectrum

Using the emulator-as-simulator NPE (§2.5) on the $C_\ell^{\kappa\kappa}$ -equivalent statistic at $N = 253$ -equivalent precision, the 30-parameter posterior conditioned on the fiducial C_ℓ alone is broad. WORKLOG’s own count of “ ~ 5 ” parameters (also the number used in the abstract) names four with an explicit posterior- σ /prior- σ ratio: IMF slope 0.66, wind energy 0.72, variable wind velocity 0.75, black-hole radiative efficiency 0.83. The corner-plot figure independently extracted from the same analysis (Figure 7) selects and displays *six* parameters as its “best-constrained” set, adding the wind free-travel-density factor and a wind-energy-reduction parameter; visually, these two additional marginals are less sharply peaked (one is edge-pinned rather than interior-peaked) than the four named above, but we do not have a sourced quantitative posterior- σ /prior- σ ratio for either of them, so we cannot state with confidence whether “4,” “ ~ 5 ,” or “6” is the more defensible count

— the mined sources themselves are not mutually consistent on this point. [TODO: Obtain the posterior- σ /prior- σ ratios for the wind free-travel-density factor and wind-energy-reduction parameter to settle whether 4, 5, or 6 parameters are appreciably informed.] The remaining ~ 24 – 26 of 30 parameters (depending on which count is adopted) are left near-prior.

This broad 30-parameter posterior is consistent with the information content of C_ℓ itself being low-rank: PCA on the full tomographic mode vector (all 510 multipole bins $\times$ 5 source planes) compresses to three components explaining 99.5% of the variance (0.863, 0.126, 0.006), with the second component representing a genuine high- ℓ direction that the earlier, coarser $\ell < 5000$ binning missed. Despite the 30-parameter marginals staying broad, the same NPE posterior projected onto the 2-D suppression latent of §3.1 is tight, occupying only ~ 0.17 – 0.18 of the width of the full Sobol-node prior-predictive cloud (Figure 8b): the auto- C_ℓ pins the suppression latent hard even though it cannot pin individual parameters.

We tested whether a finer multipole summary or full tomography improves on this. Comparing three configurations — (A) the original 10-bin, $\ell < 5000$ summary; (B) all modes out to the map Nyquist frequency ($\ell \approx 36864$, i.e. π/pixel for the 5° , 1024-pixel κ maps) compressed with PCA; and (C) all modes with no PCA compression (2550 features on 216 simulations) — all three reach $\sim 90\%$ coverage, but configuration B gives the tightest mean posterior shrink (0.986), beating both the coarse binning (0.994, A) and the uncompressed high-dimensional summary (1.004, C, which is diluted by the small-sample/high-dimensionality ratio). The suppression $S(\ell) = C_\ell/C_{\ell,\text{DMO}}$ cancels the CIC-aliasing artifact that otherwise contaminates raw C_ℓ at high ℓ , so the high- ℓ range is usable, and the run-to-run baryon signal in $S(\ell)$ actually *grows* with ℓ , from ~ 0.13 at $\ell \sim 3000$ to ~ 0.6 at $\ell \sim 45000$. Adding full tomography (all 15 unique auto+cross tomographic spectra) does not improve further: the mean posterior shrink is 0.992, slightly worse than the single-plane, all-modes result (0.986), and the tomographic mode vector still compresses to three PCA components with an almost identical split (0.864/0.125/0.006) — the cross-tomographic spectra are redundant with the auto-spectra for the baryon response at fixed cosmology.

We can diagnose *why* the 30-parameter inverse problem is so degenerate in a model-free way. The raw tomographic-mode-vector PCA described above in this section (not the $S(\ell)$ -suppression PCA of §3.1, which gives the numerically different 88.4%/10.3% split) already shows the forward map compresses to two components (86.3% + 12.6% = 98.9% of the run-to-run C_ℓ variation), and several distinct parameters drive the *same* two directions in the forward sense (forward $|\text{corr}| = 0.37$ – 0.45 for the wind-velocity factor, black-hole radiative efficiency, wind free-travel density factor, and IMF slope). Consistent with this, the 40 Sobol runs nearest the measured fiducial in C_ℓ -space span essentially the *entire* 30-dimensional prior volume (nearest-neighbor standard-deviation/prior ratio ≈ 1.01), and a linear inverse regression from C_ℓ back to individual parameters achieves $R^2 \leq 0.43$ (mean 0.06) across the 30 parameters. This is a model-free demonstration that the $C_\ell \rightarrow \theta$ inverse problem is intrinsically rank-2 (or rank-3, allowing for the high- ℓ direction above), not merely under-sampled.

Simulation-based calibration of the emulator-based NPE gives a mean 68% rank coverage of 0.70 across the 30 parameters, with a rank histogram that is approximately uniform (Figure 9b,c). An independent explicit-likelihood `emcee` cross-check was also run for this posterior (§2.5; WORK-LOG 2026-06-29 §3), and Figure 9a now visually substantiates it: the dashed `emcee` marginals overlay the solid NPE marginals for all three plotted parameters, in general agreement. (An earlier rendering of this panel had silently dropped the dashed overlay — traced during figure regeneration to a matplotlib quirk where `ax.hist(..., histtype="step", ls="--")` renders solid unless the unabbreviated `linestyle="--"` kwarg is used — and is now fixed.)

Cautionary result: stacking statistics of unequal emulator fidelity degrades the fit.

A natural next step is to combine the auto- C_ℓ with higher-order statistics and the $\kappa \times y$ cross-spectrum in the same NPE likelihood, in the hope of tightening the posterior further. Instead, stacking emulated peak counts and then $\kappa \times y$ onto the C_ℓ likelihood *degrades* the fit to the measured fiducial: the mean posterior shrink goes $0.97 \rightarrow 1.01 \rightarrow 1.03$ as we move from C_ℓ alone, to $C_\ell + \text{peaks}$, to $C_\ell + \text{peaks} + \kappa \times y$, and the number of appreciably informed parameters drops from five to zero (Figure 10). The cause is diagnostic rather than physical: the measured fiducial is comfortably in-distribution for the C_ℓ -only emulator (maximum $|z| = 2.1\sigma$ across features), but becomes a 5.2σ outlier once the emulated higher-order statistics and $\kappa \times y$ are appended, i.e. the GP emulator’s fidelity for those statistics is measurably worse than for $S(\ell)$ itself. We stress that this is a *methods* finding about the emulator, not a physics finding about the information content of peaks or $\kappa \times y$: the model-free Spearman analysis of §3.3 and the field-level $\kappa \times y$ analysis of §3.5 both show these statistics respond strongly to feedback when evaluated without relying on the emulator for the extra statistics. Any stacked-statistic SBI result built on this emulator generation should be treated with this caveat in mind.

3.5 $\kappa \times y$ and tSZ dominate field-level feedback information

A field-level, emulator-free multiprobe analysis of $N = 123$ Sobol runs (the engine docstrings hard-code this dataset size; Table 1, row †) finds the $\kappa \times y$ cross-spectrum has a large Sobol-run-to-run spread, 0.76, roughly an order of magnitude larger than the spread in WL peak counts or WST coefficients (~ 0.08). This “spread” metric is not the same quantity as the band-integrated feedback response D ranked in Figure 11a below (a separate, $N = 253$ measurement from `paper_ksz_field.ipynb`), so the two should not be read as a single consistent ranking; see the discussion after Figure 11 for how we reconcile the two. Unless otherwise noted, all further numbers in this subsection (the variance-pinning and amplitude-anchoring results below) are from this same $N = 123$ analysis. Consistently, WL κ auto-statistics (both $S(\ell)$ and peak counts) barely constrain feedback on their own, whereas the $\kappa \times y$ cross and the tSZ y auto-spectrum dominate: together they pin two feedback directions to variance 0.06 and 0.11 at fixed cosmology (Figure 11). Because $\kappa \times y$ is amplitude-dominated it is degenerate with the lensing/pressure amplitude (equivalently σ_8/Ω_m at fixed BIND cosmology), so an explicit amplitude nuisance parameter A is required to isolate the feedback direction; once A is marginalized, the amplitude-marginalized shear $\times y$ constraint on the feedback direction is comparatively modest (variance 0.25). The $\kappa \times y$ cross is largely redundant with the per-halo tSZ y signal, since both trace thermal pressure directly, so folding in both adds little further constraint; by contrast, a per-halo κ stack is a *feedback-blind* mass anchor — the per-halo lensing signal is dominated by total halo mass, and the feedback information genuinely lives at the field level.

We note that Figure 11a’s own band-integrated D ranking places $\kappa \times y$ fourth of five probes ($y\tau > \tau\tau > yy > \kappa \times y \gg \kappa\kappa$), below the $y\tau$, $\tau\tau$, and yy spectra; we do not claim $\kappa \times y$ is the single most feedback-sensitive probe in the suite, only that it is, together with the tSZ y auto-spectrum, far more sensitive than the WL κ auto-spectrum that field-level and C_ℓ auto-only SBI (§3.4) rely on.

An amplitude-anchoring exercise using the κ -auto spectrum (which scales as A^2) to pin the lensing amplitude, jointly fit with $\kappa \times y$ (which scales as A^1), tightens the amplitude constraint from $A = 1.00 \pm 0.06$ to 0.999 ± 0.008 and correspondingly tightens the feedback-direction variance from 0.25 to 0.15 — moving toward, but not reaching, the fixed-cosmology floor of 0.06 found above.

3.6 Halo-level anchor and the actual lightcone halo population

Measured directly from the fiducial composite lightcone slabs (no ray-trace randomization), the per-halo Born-approximation lensing perturbation, $\delta\kappa_{\text{halo}}(z; z_s) \sim W(\chi; \chi_s) \cdot \langle \Sigma_{\text{BIND}} - \Sigma_{\text{DMO}} \rangle_{\text{aperture}}$ with aperture $0.659 \times R_{200}$ ($\approx R_{500c}$), correlates with halo gas fraction inside r_{500c} at $r = +0.62$ (and similarly with halo temperature), with the correlation peaking near $z \approx 0.33$ (Figure 12, right panel).

An earlier version of this analysis used only the $z = 0.034$ shell as a proxy for the full lightcone-weighted halo population; we corrected this in §6 of the paper figure notebook and quantify the impact here. The $z = 0.034$ shell carries only $\sim 3\%$ of the total $z_s = 1$ lensing kernel weight, since the kernel $W(\chi)$ peaks near $z \approx 0.42$ (Figure 12, left panel). Despite this small weight, run-ranking of *amplitude* features (f_{gas} , stellar fraction, Y , mass-weighted temperature, mass-weighted entropy, evaluated at the cluster/group mass bins) at $z = 0.034$ matches the full lightcone-kernel-weighted ranking at $r > 0.95$, showing that feedback acts coherently across cosmic time; by contrast, *redistribution* features (the gas concentration ratio f_{500}/f_{200} and the ejected gas fraction $1 - f_{500}/f_{200}$) are not well proxied by the low-redshift snapshot, with $r \approx 0.47$ (Figure 12, middle panel). Using the lightcone-kernel-weighted readout instead of the $z = 0.034$ -only proxy improves the cross-validated R^2 for the radial/ejection latent direction from 0.94 to 0.95. This population is built from 253 Sobol runs plus the fiducial, each with $\sim 34,000$ halos across 20 lightcone shells spanning $z = 0.034\text{--}2.444$.

We emphasize a limitation of this per-halo lensing analysis: the Sobol suite itself has no per-run lensplanes (only randomized ray-traced κ maps), so the per-halo Born $\delta\kappa$ link above is necessarily computed on the *fiducial* realization only, not propagated across the 30-parameter prior. This is one of our mandatory caveats (§4).

3.7 First real-data contact: BIND shear $\times y$ versus DES Y3 $\times$ ACT

As a first direct confrontation with real data, we compare BIND shear $\times y$ predictions to the public DES Y3 $\times$ ACT compton- $y \times$ shear cross-correlation (Pandey/Gatti+; Gatti et al., 2022b; Pandey et al., 2022; data product `shivampcosmo/ACTxDES Y3 DES_ACT.fits`), a joint-covariance measurement reported at a total detection significance of 21σ , the highest significance to date for a shear $\times$ tSZ cross-correlation (Gatti et al., 2022b). This corrects an in-repo docstring value of “ 26.5σ ” (`desact_sheary_realfit.py`) that we could not verify against the published significance and that appears to be stale or to refer to an unspecified subset/combination; we adopt the published 21σ figure instead. In building this comparison we identified and fixed a genuine pipeline bug: the Pylians `XPk_plane` cross-power normalization differs from `Pk_plane` by a constant factor $F \approx 1.2016 \times 10^7$, which had made the stored $\kappa \times y$ cross-spectrum $\sim 10^7$ too low (auto-spectra were unaffected, and the pixel-space correlation $\text{corr}(\kappa, y) = 0.52$ was already correct); we patched this in `bind.inference.stats.power_spectrum`.

The real-data comparison itself went through a documented self-correction. An initial reading of “BIND sits $\sim 2\times$ above the data,” interpreted at the time as feedback being too weak, was subsequently identified as a pipeline error: the comparison had omitted the SZ beam entirely (BIND’s y -map is beam-free, but the real data have an effective beam) and applied no angular scale cut. A first attempt to fix this, using a $10'$ “effective” Gaussian beam with an $8'$ scale cut, found that BIND matched the data’s intermediate-angle peak but over-predicted large angles by $\sim 1.5\text{--}2\times$, with the residual entangled with cosmology, beam, scale cuts, and the $\kappa \times y$ normalization — at that stage we could draw “no clean feedback claim” from shear $\times y$. A later, more careful pass identified a further bug in this intermediate step: the $10'$ “effective” beam over-smoothed BIND, shifting its

predicted peak to $\sim 11'$ against the data's $\sim 4.5'$ peak. Replacing this with the physical $2.4'$ ACT beam gives our final, adopted result: BIND over-predicts the amplitude (IllustrisTNG feedback is too gas-bound) and is additionally too steep in angle, consistent with over-bound gas plus the finite (5°) lightcone field of view removing low- ℓ power at large angular separations (Figure 13). WORKLOG's own round-number characterization of this over-prediction is $\sim 1.5\text{--}2\times$; reading the ratios directly and more precisely off the robust, aperture-limited $8'\text{--}40'$ range of the figure itself gives BIND/data $\approx 2.5\times$ for the $\bar{z} \approx 0.74$ DES source bin and $\approx 2.3\times$ for the $\bar{z} \approx 0.94$ bin — modestly above the rounder $1.5\text{--}2\times$ figure. We adopt the figure's more precise, directly-read $2.3\text{--}2.5\times$ as the headline number in the robust angular range and treat WORKLOG's $1.5\text{--}2\times$ as the original, rounder qualitative characterization from which it was refined; we have not identified a θ -range or normalization choice in the mined sources that reconciles the two exactly, so we flag the modest ($\sim 0.3\text{--}0.5\times$) residual difference rather than silently picking one. [TODO: Confirm whether $1.5\text{--}2\times$ or $2.3\text{--}2.5\times$ is the intended final headline number by re-checking the angular range used for each.]

We stress that the un-ambiguous, systematics-robust feedback constraint at this stage of the multiprobe program still comes from the inner-CGM legs (kSZ-CAP stacking, X-ray f_{gas} ; Paper IV territory), not from shear $\times y$, precisely because of the beam/normalization/scale-cut sensitivities documented above.

4 Discussion

4.1 What “2-D” means for feedback marginalization

The result that the WL suppression response is 2-dimensional (§3.1) is not merely a data-compression convenience: it is consistent with, and provides the underlying mechanism for, the companion finding of this suite that exactly two data-driven baryon templates are necessary and sufficient to remove the TNG WL baryonic bias from a mock LSST-Y10 cosmic-shear analysis, whereas a single template is insufficient and additional templates beyond two are pure marginalization cost (Lee, 2026b, Paper II, in prep). Because the same two directions (a BH/AGN axis and a SN/wind axis) organize essentially every field statistic we tested (§3.2), a nuisance model built to marginalize the WL two-point function should, to leading order, also marginalize the higher-order and cross-correlation feedback signature — though the rotation of the second axis between probes (§3.2) means a marginalization scheme calibrated on C_ℓ alone is not guaranteed to be safe for e.g. Minkowski functionals, which see a substantially rotated combination of the same underlying physics.

4.2 Emulator fidelity as the current SBI bottleneck

The cautionary stacking result of §3.4 points to a concrete methods bottleneck rather than a physics limitation. The `bind.emulator` Gaussian-process surrogate recovers the WL suppression response to a median fractional error of 2.0% on $S(\ell)$ (dip-correlation 0.87) on a 237-run interior held-out split, comfortably below the $\sim 6\%$ cosmic-variance/BIND-generative-noise floor — good enough to support the C_ℓ -only NPE of §3.4, but evidently not yet good enough to support the stacked higher-order/ $\kappa \times y$ NPE, which produced the 5.2σ outlier described above. A complementary, quantitative Stage-IV-precision check compares a small-feature-count emulator's held-out fractional error on $S(\ell)$ ($\ell > 300$) to the LSST-Y10/Euclid measurement-error floor across sweeps of the number of physical input features: median error is 5.3% at two features and saturates at 4.4% for three, four, or five features, indicating the emulator's residual error at small scales is limited by which physical

features it is given, not by the number of training simulations. Taken together, these two results suggest that improving the emulator’s higher-order-statistic and cross-spectrum fidelity — rather than collecting more Sobol simulations per se — is the most direct route to a tighter, trustworthy stacked-statistic SBI posterior in future work. [TODO: A quantitative re-run of the stacked SBI analysis with an improved higher-order-statistic emulator is left to future work.]

4.3 TNG-only, fixed-cosmology, and interpolation-only caveats

Every result in this paper is drawn from a single hydrodynamic simulation suite (IllustrisTNG/SB35) at fixed cosmology; we have not tested whether the 2-D latent geometry, its physical axes, or the SBI information content generalize across simulation codes/subgrid models (e.g. FLAMINGO, BAHAMAS, EAGLE-descendants; Schaye et al., 2023) or across cosmological parameters. This is stated explicitly as a caveat in the underlying analysis code and we repeat it here: cross-suite and cross-cosmology generalization is untestable with the data used in this paper. Relatedly, every SBI and latent-space claim above is interior to the $n = 253$ -point, 256-node Sobol design’s prior-predictive cloud — this is an interpolation result within the sampled astrophysical prior, not an extrapolation test, and $n = 253$ points in 30 dimensions is itself a sparse covering of that space even though the effective response dimensionality appears to be ~ 2 –3. Finally, the per-halo Born $\delta\kappa$ – f_{gas} link of §3.6 is computed on the fiducial realization only, because the 30-parameter Sobol suite itself carries no per-run lensplanes, only randomized ray-traced κ maps; propagating the per-halo lensing signature across the astrophysical prior is a natural, and currently unaddressed, extension.

4.4 A note on shared figure provenance with Paper IV

Figures 11 and 13 (§3.5–3.7) were produced by the same field-level notebook and `emulator_dataset.npz/ $\kappa \times y$` pipeline used by the sibling kSZ $\times$ X-ray analysis of this suite (Lee, 2026c, Paper IV, in prep). We include them here because the underlying multiprobe engines (`wl_sz_multiprobe.py`, `wl_sz_cosmo_anchor.py`, `wl_sheary_fit.py`, `desact_sheary_realfit.py`) belong scientifically to the WL-latent/SBI narrative of this paper, but we flag explicitly that Paper IV may show the same two figures, or a close variant, for its own kSZ-centric narrative; we defer to the cross-paper coherence pass to decide final figure ownership and avoid duplicated near-identical figures across the two papers. [TODO: Coordinate with Paper IV on final figure ownership for the $\kappa \times y$ -driver and DES $\times$ ACT confrontation figures.] The same source notebook, `paper_ksz_field.ipynb`, is also where the ~ 1 -dimensional field-response PCA discussed in §3.1 originates; that measurement is in tension with this paper’s own 2-D headline result, and we have not fully reconciled the two (see the caveat paragraph immediately preceding Figure 1). Coordination with Paper IV on figure/number ownership should therefore also resolve, or explicitly carry forward, this open dimensionality tension rather than let each paper quote a different number from the same underlying notebook family.

5 Conclusions

Using a 256-node, 30-parameter Sobol sweep of the IllustrisTNG astrophysical feedback prior painted onto shared dark-matter-only lightcones with BIND, we find:

- The WL power-spectrum suppression response is two-dimensional: two linear PCA components explain 98.7% of its variance on $N = 216$ runs (0.884 ± 0.103), confirmed independently at 99.2% on the final $N = 253$ sweep and by a β -TC-VAE, with a clean physical reading as a BH/AGN axis and a SN/wind axis. This independently reproduces the two-dimensional

feedback latent found for the 3-D matter power spectrum across CAMELS suites (Lin et al., 2026).

- This latent space is nearly shared with the kSZ gas latent on its leading axis (canonical correlation 0.98, principal angle 12°) but rotated on the second axis (0.79, 38°): WL sees a blend of gas zones that kSZ separates.
- A model-free Spearman response map identifies the IMF slope, wind velocity, black-hole radiative efficiency, and wind energy as the dominant drivers of the WL suppression at $n = 253$.
- SBI on the WL auto-spectrum alone leaves individual astrophysical parameters weakly constrained but pins the 2-D suppression latent tightly ($\sim 0.17\text{--}0.18\times$ the prior-cloud width) and passes SBC calibration (mean 68% coverage 0.70); naively stacking higher-order statistics of lower emulator fidelity onto this likelihood degrades rather than improves the fit, a methods caveat rather than a physics result.
- Field-level, emulator-free evidence shows the $\kappa \times y$ cross-spectrum and the tSZ auto-spectrum, not the WL auto-spectrum, carry the bulk of the feedback information (pinning two directions to variance 0.06/0.11), directly explaining why C_ℓ -only SBI is broad; a per-halo κ stack is a feedback-blind mass anchor.
- Using the actual lightcone halo population rather than a single low-redshift proxy snapshot, per-halo Born $\delta\kappa$ tracks halo gas fraction at $r = +0.62$; the low-redshift proxy is a good stand-in for feedback *amplitude* ($r > 0.95$) but a poor one for gas *redistribution* ($r \approx 0.47$).
- The first direct confrontation of BIND shear $\times y$ with real DES Y3 $\times$ ACT data finds BIND too high (WORKLOG’s rounded characterization: $\sim 1.5\text{--}2\times$; the precise robust-aperture readout: $\approx 2.3\text{--}2.5\times$, §3.7) and too steep at the TNG fiducial point, i.e. IllustrisTNG feedback is too gas-bound relative to the real Universe at this level of analysis, consistent in direction with independent inner-CGM constraints.

[TODO: A calibrated, population-level SBI posterior on the full 8.6-million-halo per-object dataset (§2.6) was designed but not completed on GPU, and is deferred to future work rather than reported quantitatively here.] These results, together with Paper II’s independent capstone finding that two nuisance parameters suffice for LSST-Y10 marginalization (Lee, 2026b, Paper II, in prep), argue that the practically relevant dimensionality of baryonic feedback for WL analyses is $\mathcal{O}(2)$, not $\mathcal{O}(30)$ — but that the field-level statistics best able to *pin* that low-dimensional space (SZ cross-correlations) are different from the two-point WL statistics typically used to marginalize it, motivating the multiprobe direction pursued throughout this suite.

A Optional appendix: SHMR scatter budget on the SB35 Sobol suite

As a related, lower-priority result from the same Sobol suite, we measure the stellar-mass–halo-mass relation (SHMR) scatter budget at fixed halo mass. The Sobol suite, a companion “twobound” one-at-a-time feedback sweep, and the fiducial run all paint the identical 2933 snapshot-96 dark-matter halos (matched 2933/2933), so the comparison below is a genuine drop-in reuse rather than a resimulation.

Jointly varying all 30 feedback parameters produces a scatter in $\log_{10} M_\star$ at fixed halo mass of $\sigma \approx 0.386$ dex (variance ≈ 0.149 dex²), which is larger than the scatter obtained by varying each of the twobound one-at-a-time parameters individually, $\sigma \approx 0.161$ dex (variance ≈ 0.026 dex²; Figure 14). Because the joint variation compounds across parameters, this shows that single-knob extremes from a one-at-a-time sweep are *not* a valid upper bound on the true joint-feedback scatter, correcting a caveat noted in the original twobound analysis; we further note that the twobound one-at-a-time comparison line is itself a conservative marginal (not a paired per-run) error convention, so the comparison above is directionally, not exactly, quantitative. Accretion history — measured as the formation time t_{form} on the dark-matter merger tree — explains roughly 10% of the remaining halo-to-halo (non-feedback) SHMR residual, with a halo-to-halo variance of ≈ 0.005 dex² and early-forming halos sitting systematically above the mean relation.

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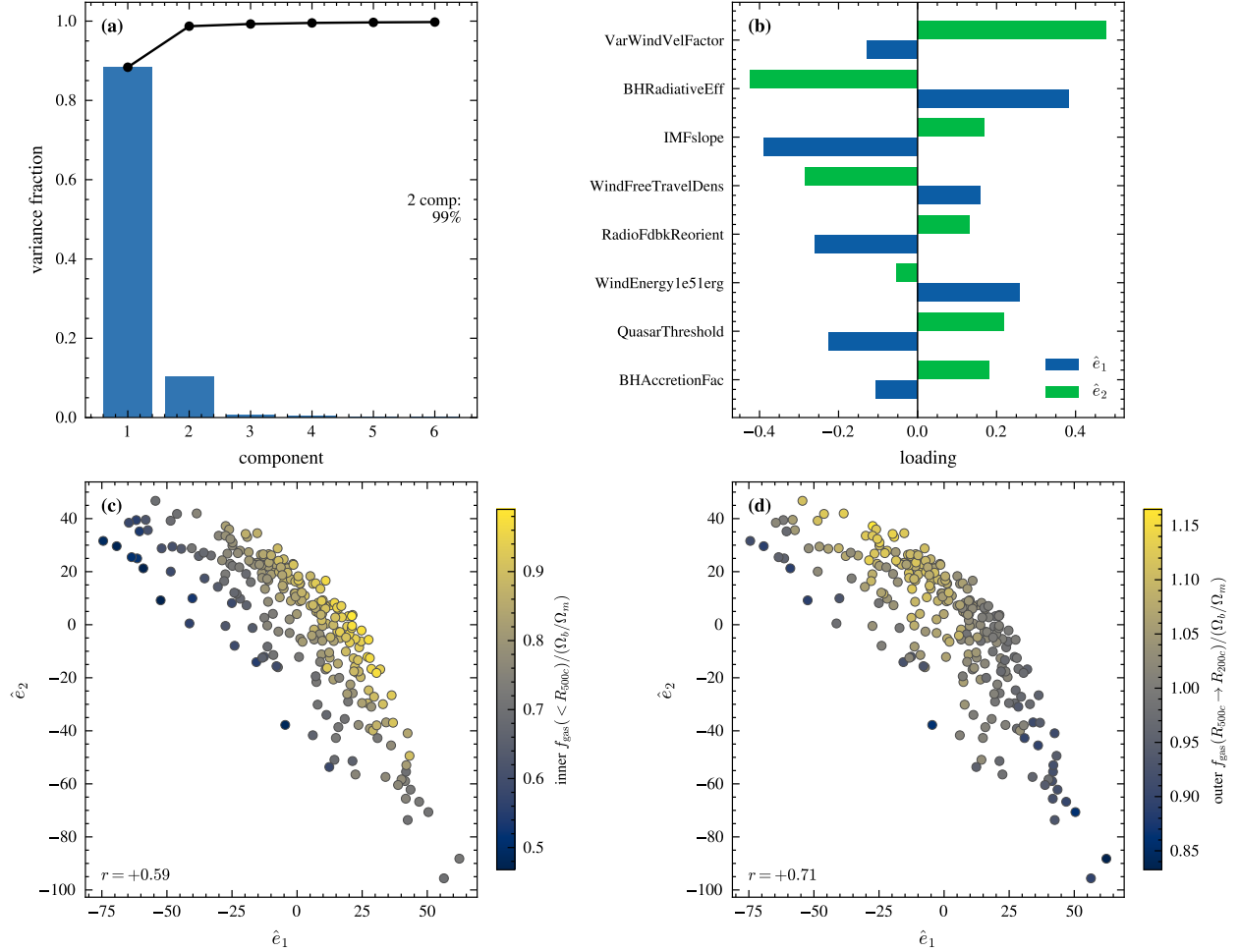


Figure 1: The 2-D latent space of the WL suppression $S(\ell, z_s)$, from the β -TC-VAE decomposition on the $N = 216$ Sobol subsample. (a) Linear scree confirms two components capture 99% of the variance. (b) The parameters with the largest $|\text{loading}|$ on each latent axis, read directly off panel (b): *IMFslope* and *BlackHoleRadiativeEfficiency* dominate $\hat{e}_1$ (nearly tied, $|\text{loading}| \approx 0.38$ – 0.39 each), while *VariableWindVelFactor* and *BlackHoleRadiativeEfficiency* dominate $\hat{e}_2$ (≈ 0.48 and ≈ 0.42). Because *BlackHoleRadiativeEfficiency* loads strongly on *both* axes, the loadings alone do not cleanly separate a single-parameter wind/SN axis from a single-parameter BH/AGN axis; the wind/SN-vs-BH/AGN physical reading is instead supported by the CCA/1P-spoke evidence of §3.2 and Figure 3. (c)/(d) The same latent plane, colored by the halo inner ($< R_{500}$) and outer ($R_{500} \rightarrow R_{200}$) gas fraction: WL tracks both zones only moderately well ($r = +0.59$ inner, $+0.71$ outer), which we contrast with kSZ’s much cleaner separation of the same two zones in §3.2. **Takeaway:** the WL feedback response is governed by two physical directions — a wind/SN sector and a BH/AGN sector — not by 30 independent knobs.

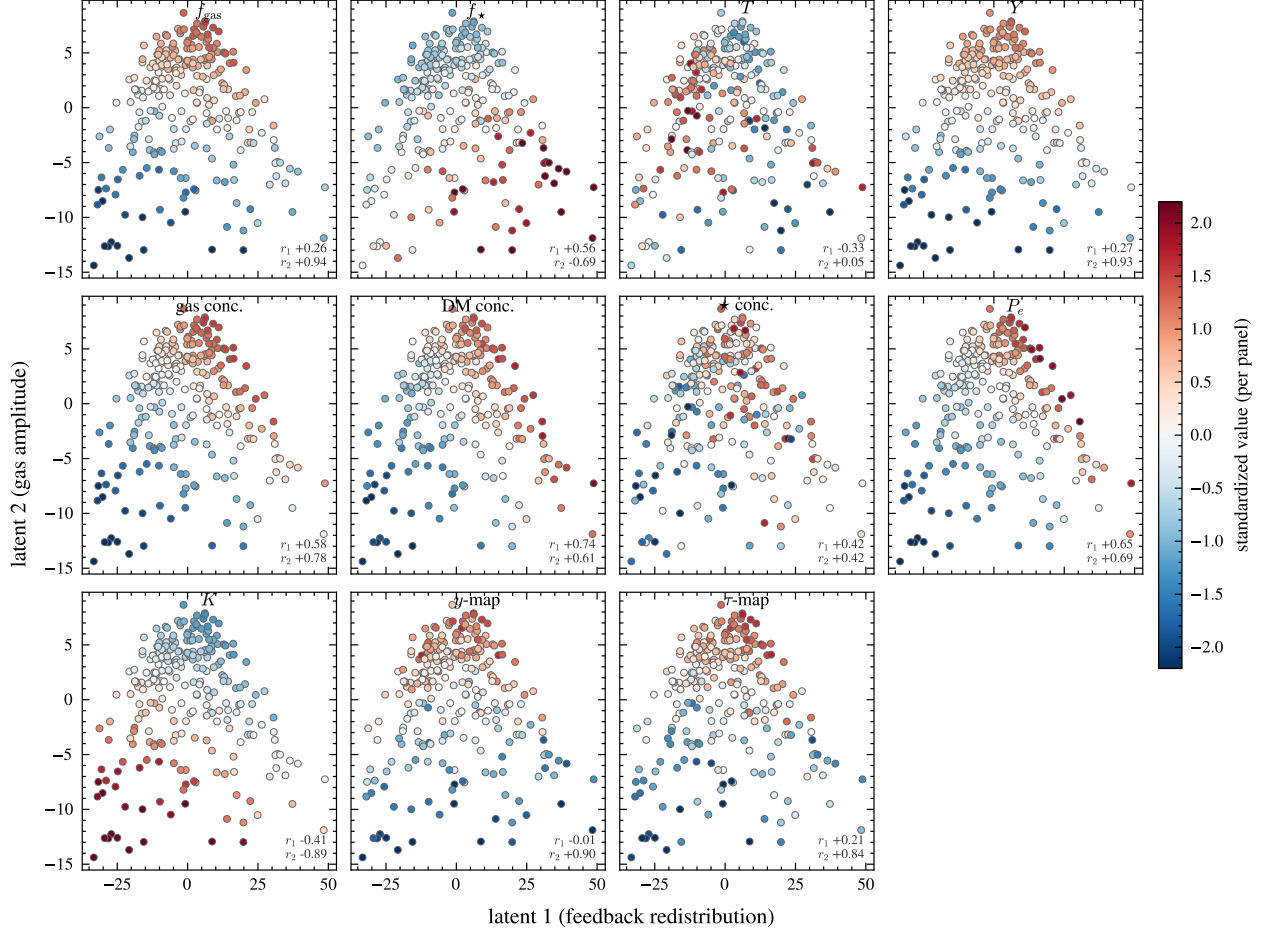


Figure 2: Physical correlates of the two latent coordinates, measured on the $N = 253$ final Sobol sweep (`paper.lightcone.figs2.ipynb` §6.1). Each panel shows the same latent plane colored by a different physical quantity, with the Pearson correlations r_1 (latent 1) and r_2 (latent 2) annotated. Latent 2 aligns most strongly with f_{gas} ($r_2 = +0.94$), integrated Compton- Y ($r_2 = +0.93$), the projected tSZ y -map ($r_2 = +0.90$), and the kSZ τ -map ($r_2 = +0.84$), while entropy K is anti-correlated with both axes ($r_1 = -0.41$, $r_2 = -0.89$); latent 1 aligns most strongly with dark-matter concentration ($r_1 = +0.74$) and electron pressure P_e ($r_1 = +0.65$), with gas concentration ($r_1 = +0.58$) and the stellar fraction ($r_1 = +0.56$) also substantial — the panel’s x -axis is labeled “ $\sim f_\star$ ” following the plot’s own naming convention, though P_e and gas concentration correlate more strongly with latent 1 in raw magnitude. **Takeaway:** latent 2 is essentially a thermal-energy/gas-content axis (it tracks f_{gas} , Y , y , τ together), confirming that the same two latent coordinates that organize the WL suppression also organize the gas thermodynamics directly.

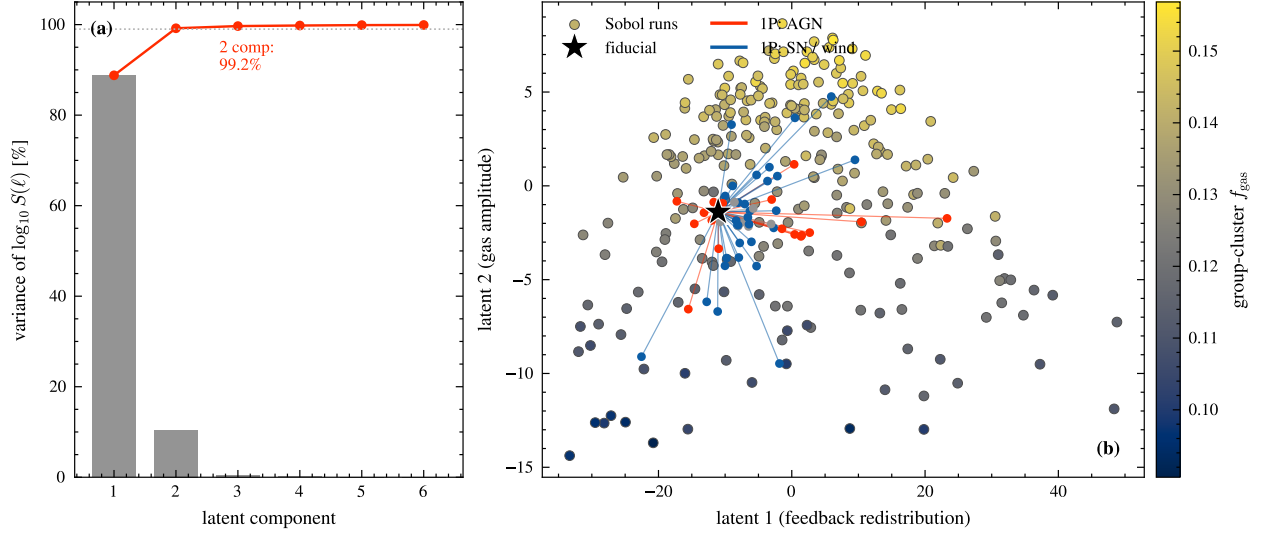


Figure 3: The independent $N = 253$ repeat of the latent-plane construction (`paper_lightcone_figs2.ipynb` §5.1). Left: linear scree showing $\text{PC1} \approx 89\%$, $\text{PC2} \approx 11\%$ (independently rounded bar-height readouts; the underlying values sum to the annotated cumulative), cumulative 99.2% — consistent with the 98.7% headline number of Figure 1a. Right: the 253 Sobol nodes (circles, colored by group-cluster f_{gas}) and the fiducial (black star) on the (latent-1, latent-2) plane, with the astrophysical one-parameter-at-a-time (1P) AGN (red) and SN/wind (blue) spokes overlaid. **Takeaway:** the 1P spokes for the AGN and SN/wind parameter families point in visibly different directions on the latent plane, providing an independent, non-Sobol confirmation that the two latent axes correspond to the two physical feedback channels identified in Figure 1.

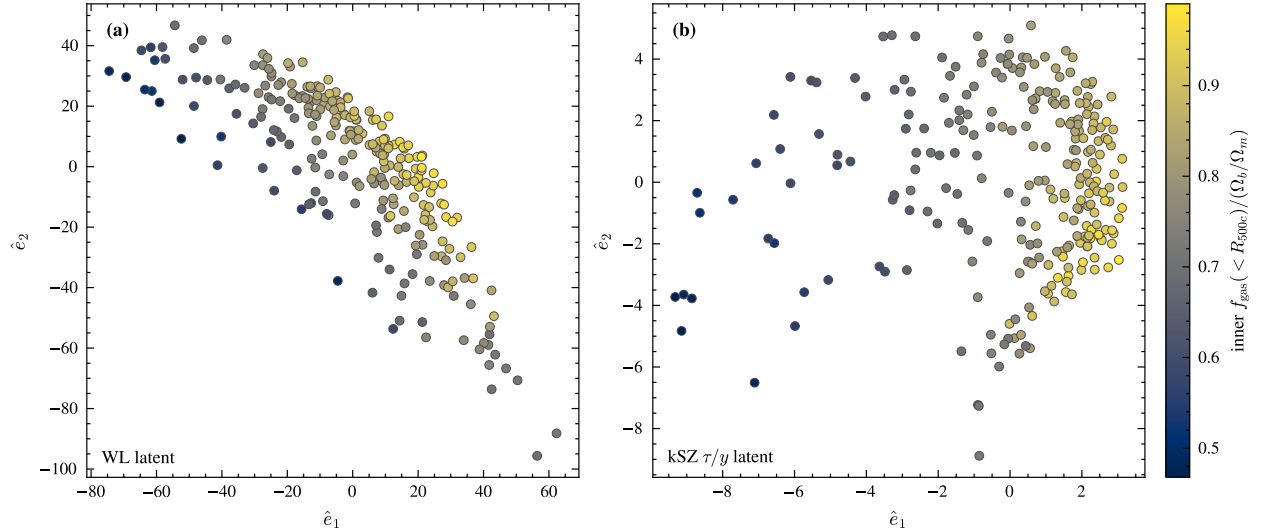


Figure 4: Canonical-correlation geometry between the WL suppression latent (left) and the kSZ τ/y latent (right), both colored by the same inner-halo gas fraction. The two planes have canonical correlations $[0.98, 0.79]$ and principal angles $[12^\circ, 38^\circ]$. **Takeaway:** the leading feedback direction is probe-independent (“same leading direction”), but the second direction is probe-dependent — WL blends gas zones that kSZ keeps separate.

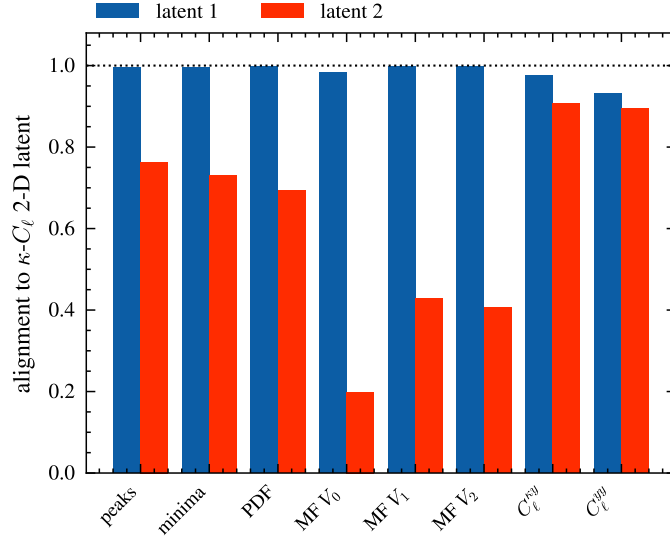


Figure 5: Alignment of each WL summary statistic’s own top-2 PCA latent to the κ - C_ℓ 2-D latent, for eight statistics. Latent 1 (blue) is nearly universal (≥ 0.93); latent 2 (red) ranges from 0.20 (Minkowski V_0) to 0.90 ($C_\ell^{\kappa y}$). **Takeaway:** every WL statistic sees the same leading feedback direction, but only the cross- and tSZ auto-spectra see the same *second* direction that kSZ sees; the higher-order moment statistics see a rotated combination.

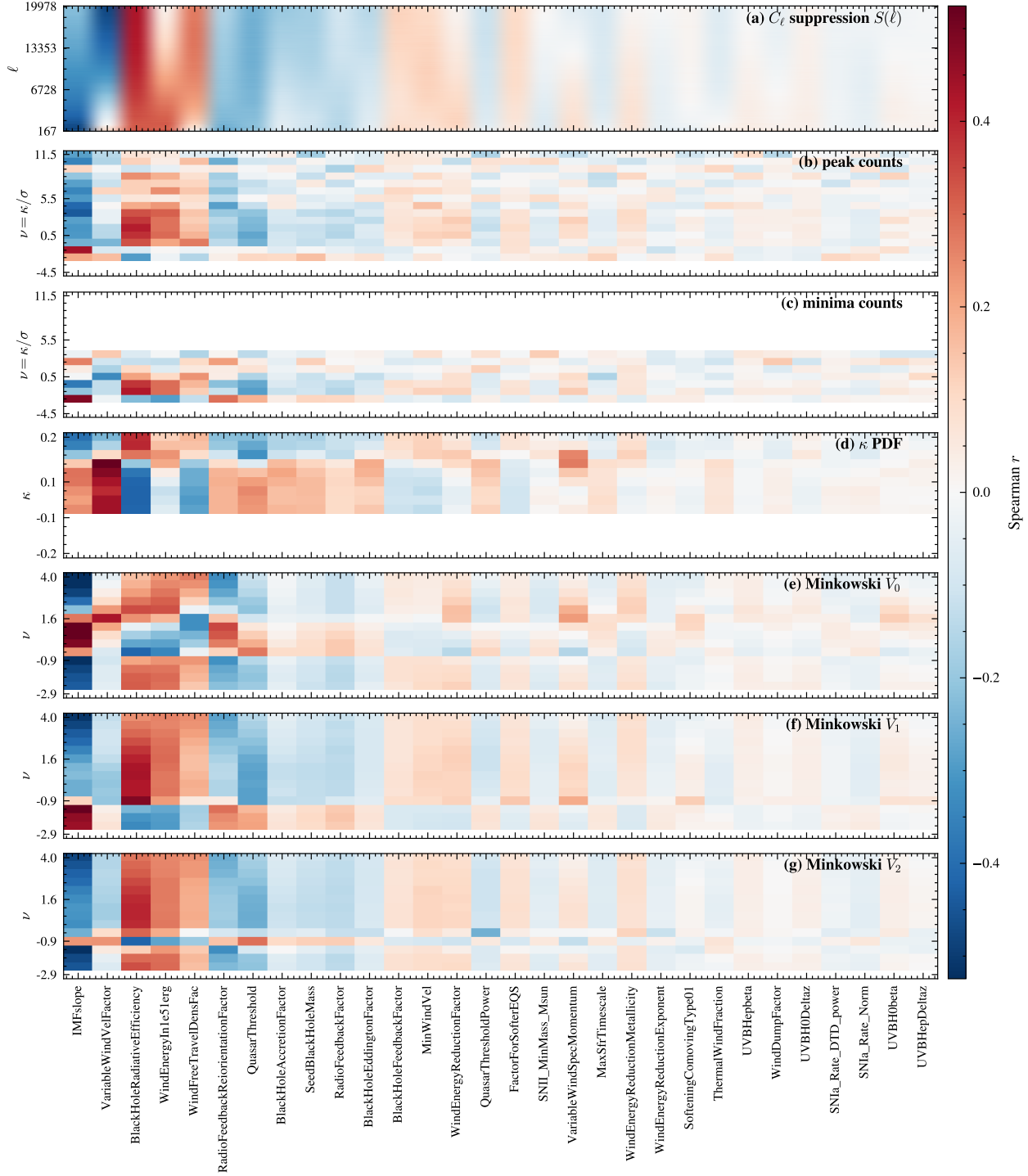


Figure 6: Model-free per-bin Spearman correlation of seven field statistics against all 30 SB35 astrophysical parameters, computed directly on the $n = 253$ measured Sobol runs at $z_s = 1$ (deliberately emulator-free). **Takeaway:** response is concentrated in a handful of wind/SN and BH parameters (left columns, dark bands), with essentially no response from the majority of the 30-dimensional prior (e.g. softening lengths, SNIa rate parameters) — direct evidence, independent of any latent-space construction, that the effective dimensionality of the response is low.

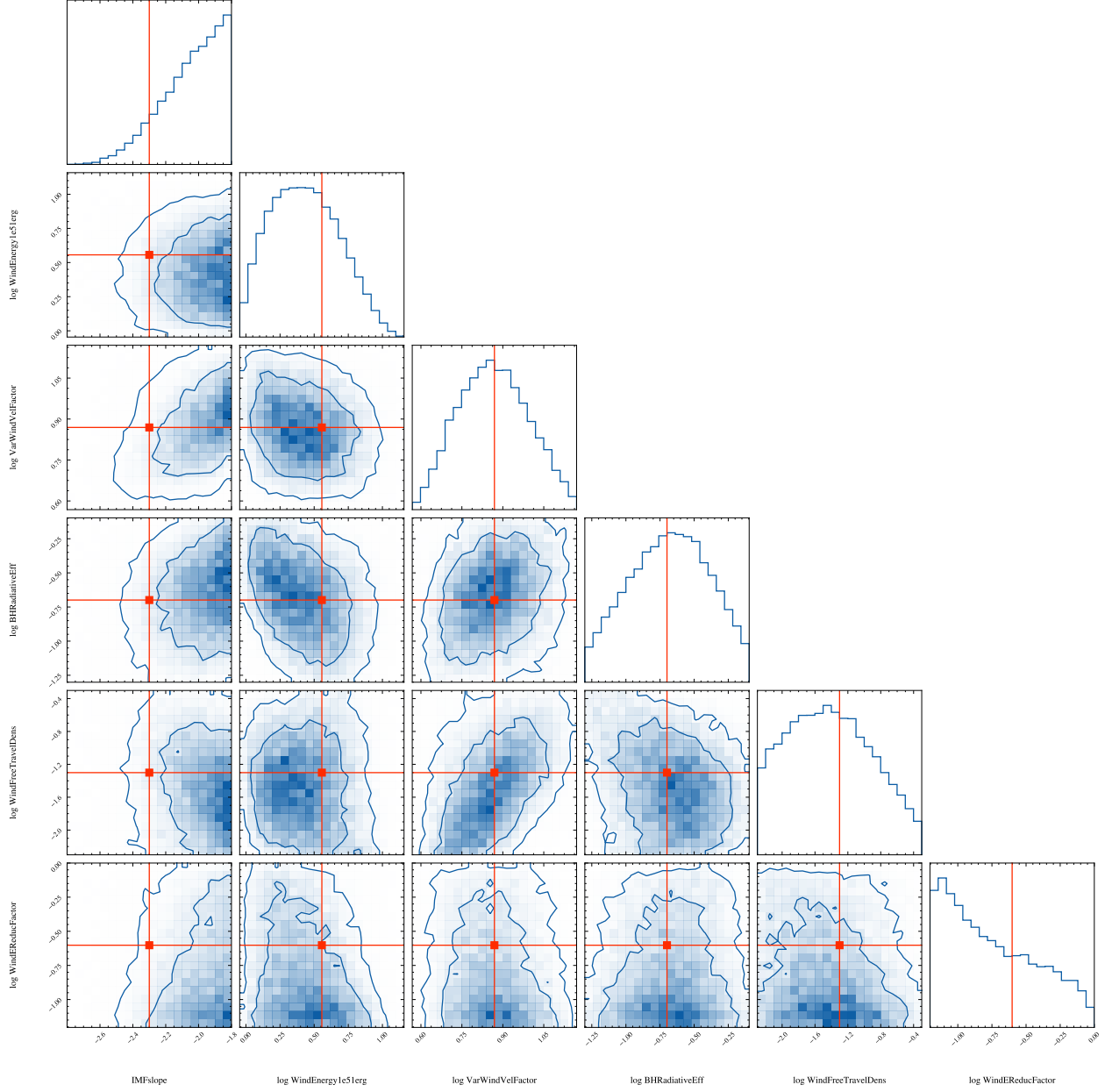
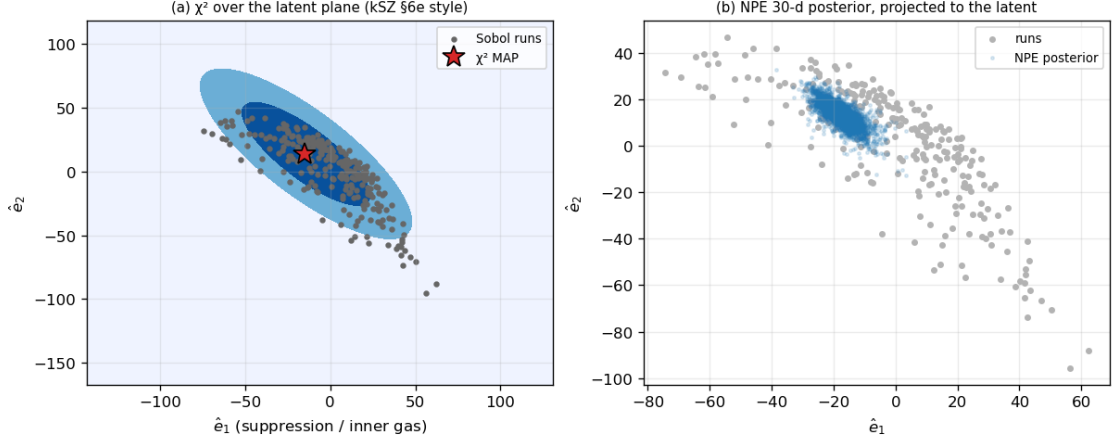


Figure 7: Neural posterior $P(\theta \mid C_\ell^{\text{fiducial}})$ for the six parameters the source notebook selects as best-constrained, from NPE against the emulator-as-simulator (§2.5); the red lines/point mark the true TNG fiducial parameter values. Four of the six (IMF slope, wind energy, variable wind velocity, black-hole radiative efficiency) have sourced posterior- σ /prior- σ ratios in the main text; the other two (wind free-travel density, wind-energy reduction) are shown here but their ratios are not available in the mined sources — see the caveat in the main text. **Takeaway:** the auto- C_ℓ constrains the wind/SN/BH sector weakly but recovers the fiducial within its 1–2 σ contours; the majority of the remaining ~ 24 –26 of 30 parameters (not shown) are essentially unconstrained.



[TODO:

regenerate fig07 — needs one CPU-only forward pass of the cached GP emulator checkpoint to build the chi2 grid / latent-projected posterior cache; not yet run (see `fig_scripts/UNREGENERATED.md`)]

Figure 8: (a) χ^2 over the 2-D suppression latent plane with the maximum-a-posteriori (MAP) point marked; the Sobol node cloud (grey) is overplotted. (b) The full 30-dimensional NPE posterior (Figure 7), projected onto the same 2-D latent (blue), compared to the Sobol node cloud (grey). **Takeaway:** even though the 30-parameter marginals are broad (Figure 7), the projection onto the physically meaningful 2-D latent is tight ($\sim 0.17\text{--}0.18\times$ the node-cloud width) — the auto- C_ℓ constrains the response, not the individual knobs.

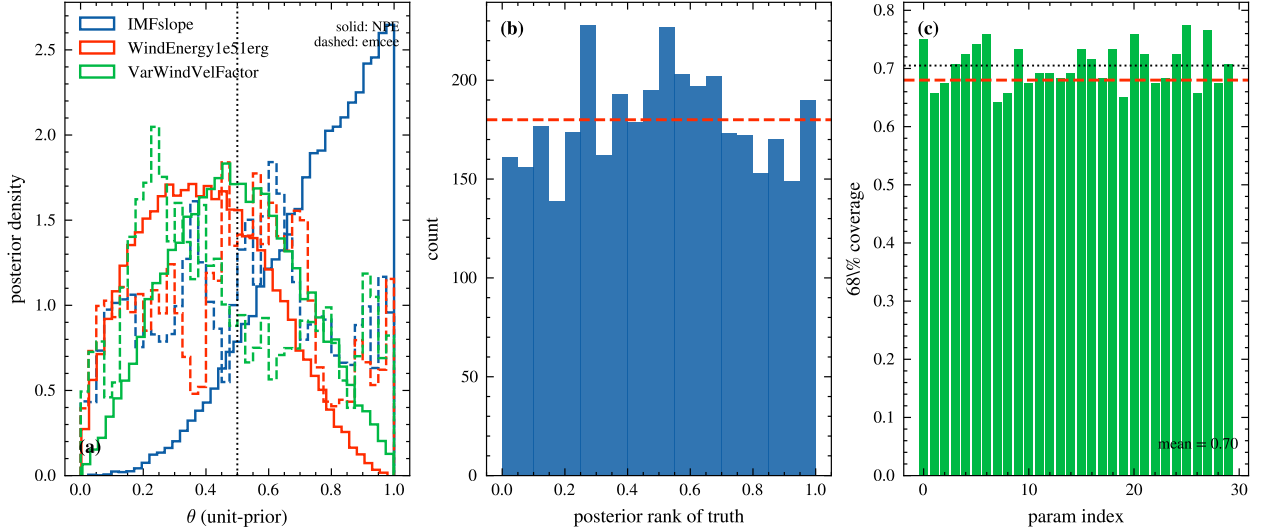


Figure 9: Cross-checks on the NPE posterior. (a) NPE posterior marginals (unit-prior units, solid) for three representative parameters, with the independent explicit-likelihood `emcee` marginals (dashed) overlaid. (b) Simulation-based calibration rank histogram, approximately uniform if the posterior is well calibrated. (c) Per-parameter empirical 68% coverage across all 30 parameters, mean = 0.70 (red dashed line). **Takeaway:** panels (b)/(c) show the NPE posterior is reasonably well calibrated (mean coverage close to the nominal 0.68), which is the basis for the broad-vs-tight contrast between the 30-parameter and 2-D-latent posteriors above; panel (a) now visually substantiates the NPE/`emcee` agreement referenced in §2.5.

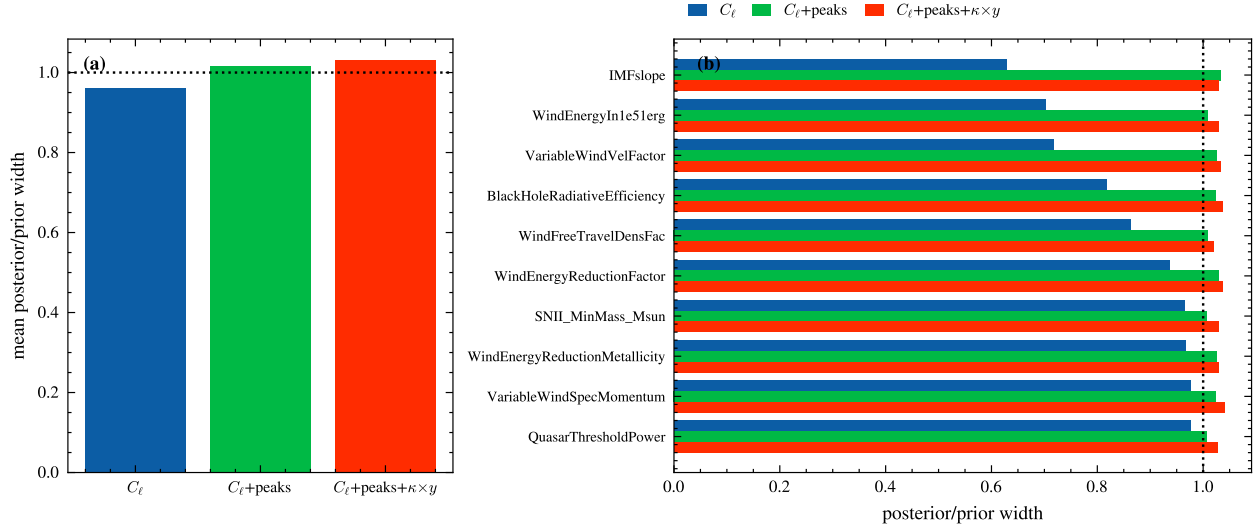


Figure 10: Left: mean NPE posterior/prior width as higher-order statistics are progressively stacked onto the C_ℓ likelihood ($C_\ell \rightarrow C_\ell + \text{peaks} \rightarrow C_\ell + \text{peaks} + \kappa \times y$); values > 1 indicate the posterior is *wider* than the prior at the measured fiducial, i.e. the fit has degraded. Right: the same comparison, per-parameter, for the ten best-constrained parameters, in order: *IMFslope*, *WindEnergyIn1e51erg*, *VariableWindVelFactor*, *BlackHoleRadiativeEfficiency*, *WindFreeTravelDensFac*, *WindEnergyReductionFactor*, *SNII_MinMass_Msun*, *WindEnergyReductionMetallicity*, *VariableWindSpecMomentum*, and *QuasarThresholdPower* — with untruncated labels, the two similarly named *WindEnergyReduction** parameters in this list are *WindEnergyReductionFactor* (rank 6) and *WindEnergyReductionMetallicity* (rank 8); the third, *WindEnergyReductionExponent*, is not among the ten. **Takeaway:** this is a warning about emulator fidelity for higher-order/cross statistics, not evidence that those statistics lack feedback information (contrast with Figures 6 and 11).

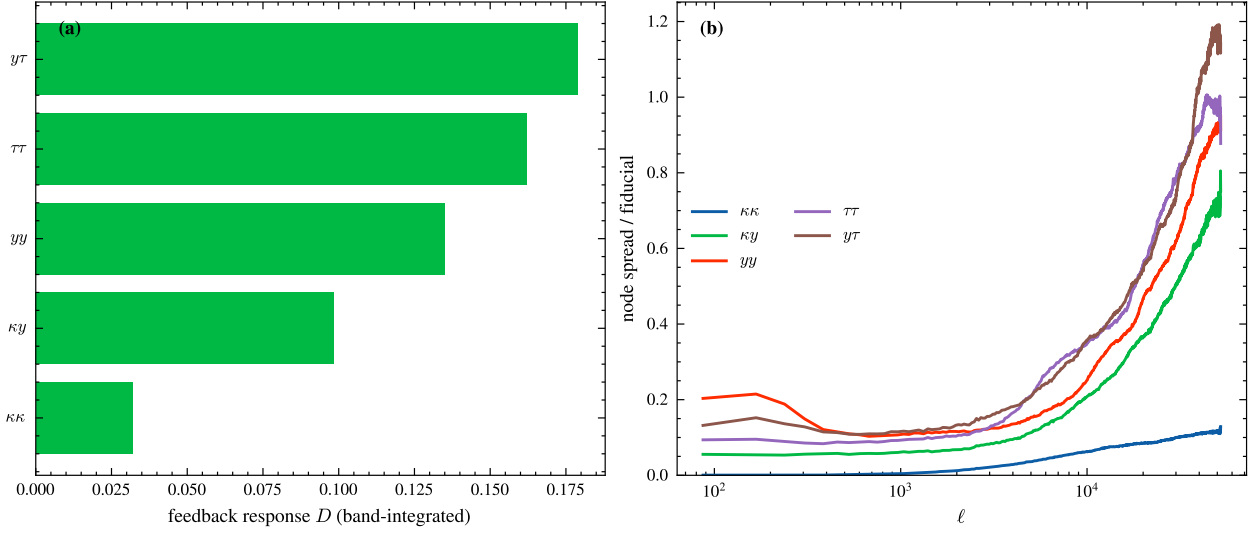


Figure 11: (a) Band-integrated feedback response D by probe, ranked $y\tau > \tau\tau > yy > \kappa \times y \gg \kappa\kappa$ (WL auto is the least feedback-sensitive probe by an order of magnitude). (b) Sobol node spread relative to the fiducial as a function of multipole for the same five probes: the WL auto-spectrum ($\kappa\kappa$, blue) stays flat and dark out to $\ell \sim 10^4$, while the cross- and tSZ/kSZ probes “light up” at far lower ℓ . This figure was produced from the same field-level machinery (`emulator_dataset.npz`, the $\kappa \times y$ cross pipeline) shared with the sibling kSZ analysis (Lee, 2026c, Paper IV, in prep); see §4 for a note on this shared provenance. **Takeaway:** WL auto-statistics alone are the *least* feedback-sensitive probe in the suite — the tSZ/kSZ cross- and auto-spectra carry the bulk of the field-level feedback information, which directly explains why the SBI posteriors of §3.4 on C_ℓ alone are broad.

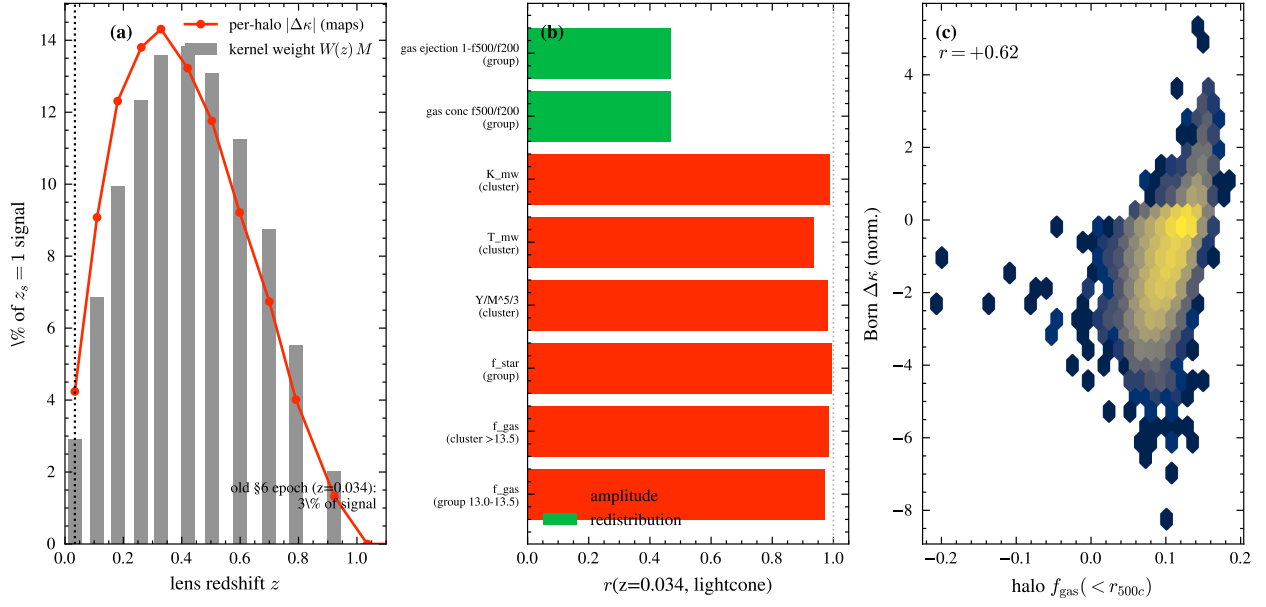


Figure 12: Left: the $z_s = 1$ lensing-kernel weight $W(z)$ (grey bars) versus lens redshift, with the per-halo $|\delta\kappa|$ signal from the actual maps overlaid (red); the old, low-redshift-only proxy epoch ($z = 0.034$) carries only 3% of the total kernel weight. Middle: correlation between the $z = 0.034$ -only readout and the full lightcone-kernel-weighted readout, separately for “amplitude” features ($r > 0.95$, red) and “redistribution” features ($r \approx 0.47$, green). Right: per-halo Born $\delta\kappa$ versus halo $f_{\text{gas}}(< r_{500c})$, $r = +0.62$. **Takeaway:** the low-redshift snapshot used in earlier work is a fair proxy for *how strong* feedback was, but not for *where the gas ended up* — the redistribution signature requires the full lightcone population, and Born $\delta\kappa$ is a genuine, though noisy, per-halo gas tracer.

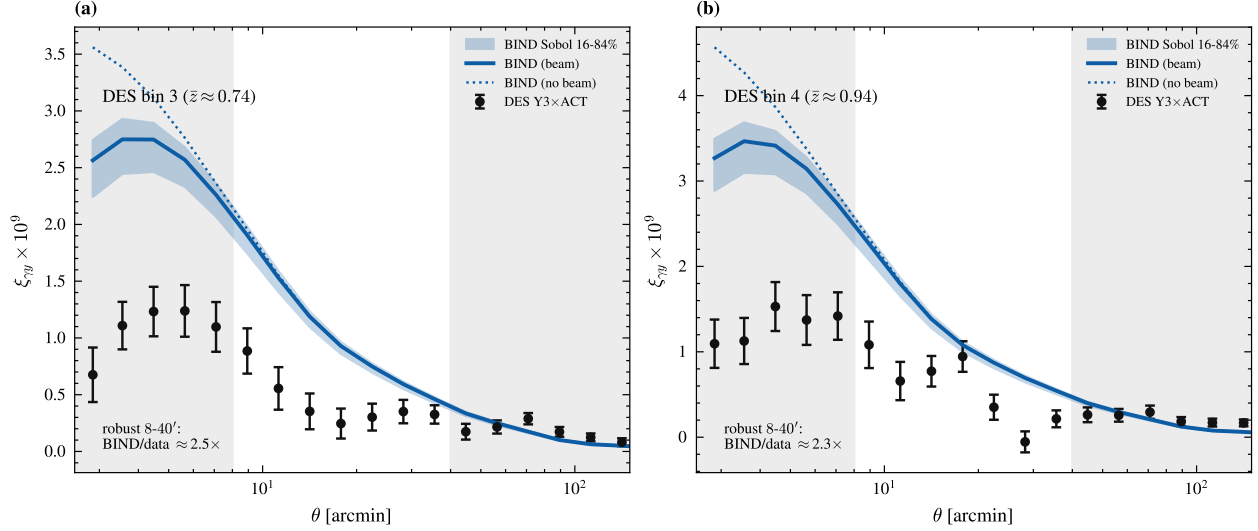


Figure 13: BIND shear $\times y$ predictions (blue; 16–84% Sobol band, solid line with the physical 2.4' ACT beam, dotted without any beam) versus the real DES Y3 $\times$ ACT measurement (black points), for two DES source-redshift bins. Grey shading marks the beam/resolution- and aperture-limited scales excluded from the robust comparison. This figure was produced from the same field-level, real-data pipeline shared with the sibling kSZ analysis (Lee, 2026c, Paper IV, in prep); see §4 for a note on this shared provenance. **Takeaway:** BIND, run at the TNG fiducial parameters, is too high and too steep relative to real DES $\times$ ACT shear $\times y$ — WORKLOG's original, rounder characterization is $\sim 1.5\text{--}2\times$, while reading the ratio directly off this figure's robust 8'–40' aperture range gives $\approx 2.3\text{--}2.5\times$ (see the main text for this not-yet-fully-reconciled difference) — the same qualitative direction (too gas-bound) reported from X-ray and kSZ-CAP f_{gas} constraints, though we stress below that the shear $\times y$ residual is currently entangled with several systematics and is not yet a clean, stand-alone feedback measurement.

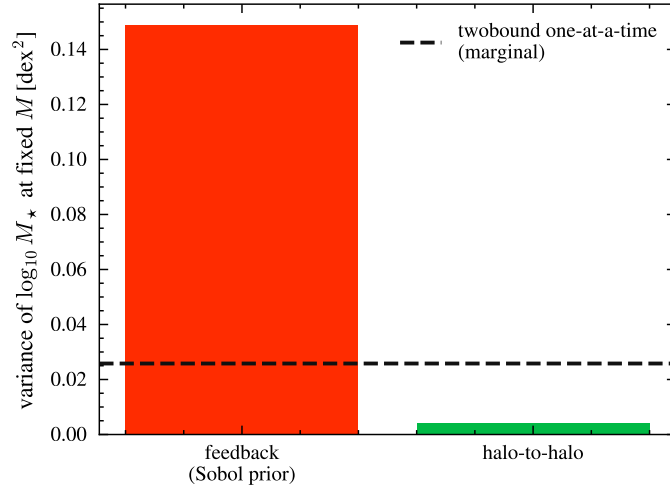


Figure 14: SHMR scatter budget on the SB35 Sobol suite: variance of $\log_{10} M_{\star}$ at fixed halo mass from jointly varying all 30 feedback parameters (“feedback,” red) versus the halo-to-halo (non-feedback) residual (green), with the twobound one-at-a-time marginal scatter overlaid as a dashed reference (black) line. **Takeaway:** joint 30-parameter feedback variation compounds well past the one-at-a-time envelope — single-knob sweeps under-estimate the true feedback-driven SHMR scatter by a factor of ~ 6 in variance.